

ISYE 6420 Homework 6

Question 1

This model will be a time to event, right censoring problem, as mice can be censored or die before true study end time. Firstly, in order to perform the analysis the missing data needed to be handled. As the missing values indicated the mice were censored, these mice were noted as the censored population to be used as the upper limit for the time to event model. These *NaN* values were filled in with the upper limit stated in the corresponding censored column. As done in the lecture notes - the values in the censored array were set to *inf* if the corresponding y value is not censored, and equal to the y value if it should be censored.

Weibull PDF:

$$f(x|\alpha, \beta) = \frac{\alpha x^{\alpha-1} e^{-(x/\beta)^\alpha}}{\beta^\alpha}$$

In order to analyse each group individually, β was created in PyMC as vector with shape=4. A new predictor, $\beta_{prime} = e^{-\beta_j}$, was also produced which will be the scale parameter for the individual groups. The priors are assumed to be non-informative and defined as such:

$$\alpha \sim \text{Gamma}(0.001, 0.001)$$

$$\beta \sim N(0, 1000)$$

β is chosen to be a normal distribution, so that the effects of photocarcinogenicity can be assessed as having a negative or positive effect. Hence β can be both positive and negative. The scale parameter used in the Weibull likelihood is exponentiated to ensure the scale parameter is >0.

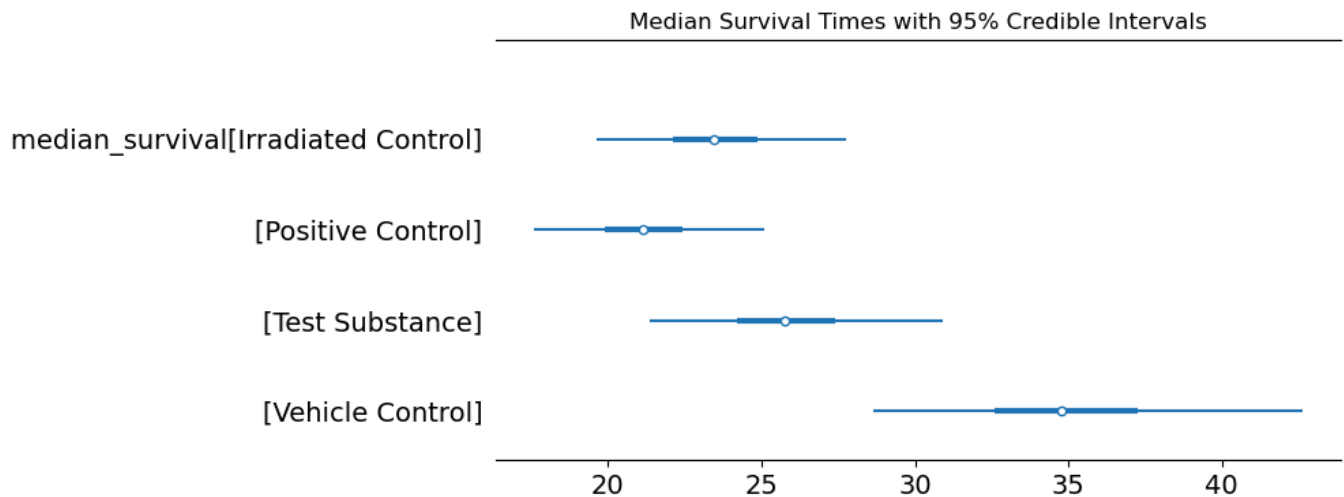
1a)

The estimates of the median survival time of each group with their associated uncertainties (95% HDI) are summarised below:

Table 1.1	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_
median_survival[Irradiated Control]	23.542	2.062	19.614	27.699	0.01	0
median_survival[Positive Control]	21.232	1.924	17.554	25.097	0.009	0
median_survival[Test Substance]	25.929	2.45	21.284	30.847	0.011	0.0

Table 1.1	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_
median_survival[Vehicle Control]	35.097	3.658	28.313	42.429	0.018	0.0

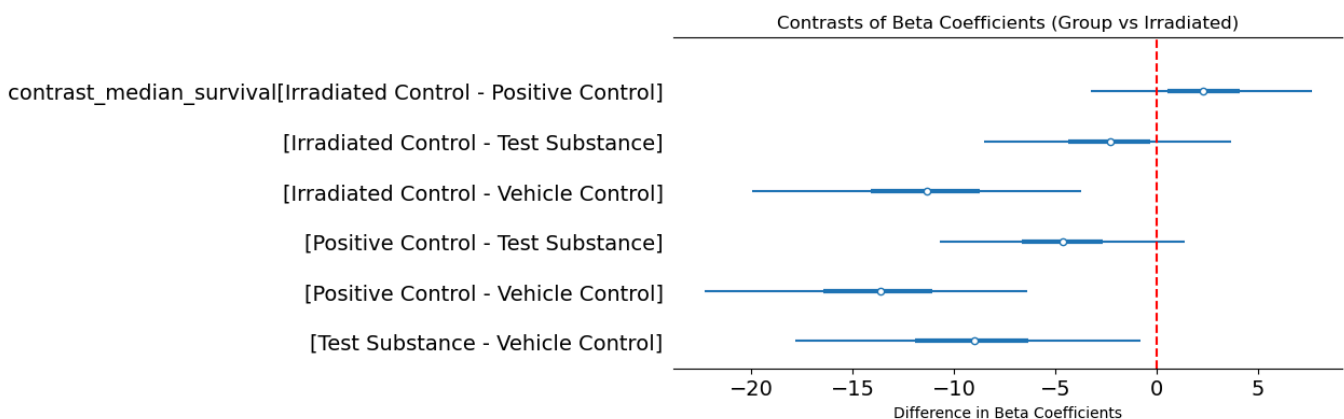
Figure 1



Based on these results of median survival times, the Vehicle Control group shows a distinctly longer survival time in comparison to the other groups. It's median value is outside the confidence intervals of the other groups.

1b)

Contrasts were calculated using the function in the lecture notes, in order to compare values between groups (particularly the significant Vehicle Control group from part a). Below are the comparisons for Median Survival Time.



1c)

Table 1.2	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
survival_at_30[Irradiated Control]	0.255	0.079	0.107	0.409	0	0

Table 1.2	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
survival_at_30[Positive Control]	0.164	0.073	0.039	0.309	0	0
survival_at_30[Test Substance]	0.348	0.09	0.179	0.528	0	0
survival_at_30[Vehicle Control]	0.631	0.081	0.469	0.783	0	0

Question 2

2a)

Posterior summaries for the predictors can be seen in Table 2.1. Note, target acceptance criteria was set to 0.95, as at the default value multiple divergences appear. The Bayesian R^2 value is ~0.76 for this model.

Table 2.1	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
alpha	9.092	9.47	-9.665	27.795	0.129	0.092
beta[0]	3.407	0.543	2.348	4.508	0.007	0.005
beta[1]	-1.456	0.244	-1.927	-0.964	0.002	0.002
beta[2]	0.328	0.272	-0.208	0.868	0.003	0.002
sigma	2.702	0.441	1.937	3.587	0.003	0.003

2b)

In order to determine outliers and goodness of fit, a cumulative distribution plot of posterior vs observed, ECDF PIT and LOO-PIT values were produced as seen below in Figure 2.1 and Table 2.3 respectively. The combination value (squared logit of LOO PIT and ECDF PIT), highlight three major outliers.

Figure 2.1

Posterior Predictive Check - Cumulative Distribution of Predicted vs Observed

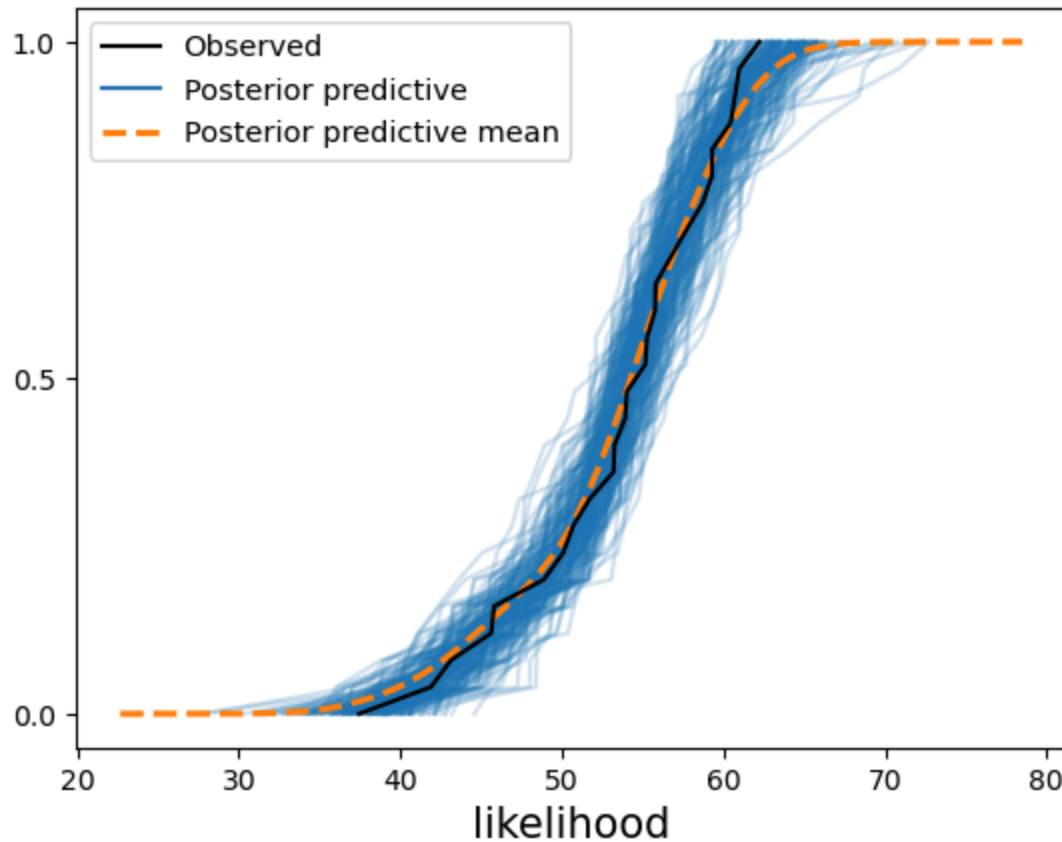


Table 2.2: PSIS LOO and ECFD PIT Top 5	likelihood	loo_pit	ecdf_pit	squared_logit
14	53.254	0.000181805	0.004975	28.0724
17	50.086	0.0129563	0.03385	11.2317
16	58.699	0.954603	0.9404	7.61015
1	59.295	0.810913	0.790475	1.76303
19	62.213	0.791031	0.767325	1.42389

2c)

Two Robust Models were built using Student's T distribution. One model had a less informative prior on ν as an exponential distribution $\sim \text{Exp}(\lambda = 1/30)$. The second model had a more informative prior where ν is a gamma distribution $\sim \text{Ga}(\alpha = 3, \beta = 0.1)$.

Using LOO-PIT to compare all 3 models produced the results in Table 2.4, showing that the Student T distribution models were improvements on the original model.

Table 2.3: PSIS- LOO CV for 3 models	rank	elpd_loo	p_loo	elpd_diff	weight	se	dse
Robust_v1	0	-51.9369	7.74675	0	0.974857	8.5493	0
Robust_v2	1	-55.2845	9.67296	3.34756	0	9.04419	1.03681
Original	2	-65.9289	6.15476	13.992	0.0251428	6.94094	4.64862

2d)

Prediction at new observation points:

Table 2.4	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
mu_response[0]	28.006	5.026	17.235	36.975	0.1	0.196
predictive_response[0]	30.591	51.405	11.494	38.796	2.819	24.178

Question 3

A linear regression model was fit with the provided priors in the question. As we are assuming MCAR, the missing variables in the outcome can be handled directly in the PyMC model through imputation from the sampling distribution.

Table 3.1	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
alpha	614.227	11.003	592.493	635.825	0.109	0.076
beta[Hardness]	60.436	9.198	42.326	78.55	0.074	0.049
beta[Rake]	-4.015	8.68	-21.219	13.138	0.058	0.048
beta[Helix]	-158.225	9.734	-177.262	-138.871	0.078	0.052
beta[Relief]	-9.598	8.628	-26.654	7.123	0.062	0.046
beta[Corner]	-4.684	8.509	-21.972	11.468	0.061	0.045
beta[Flute]	84.449	9.525	65.98	103.477	0.067	0.052
sigma	17.991	2.197	14.016	22.386	0.016	0.012

3b)

When considering MNAR, the missing values in the outcome were masked and set to -1 instead of NaN in order for the PyMC model to run without issues. When building the missingness model, the 50% breakage chance at Force=650, will be the midpoint of the

distribution - in the logscale this will be centred at 0 and the data points will be in reference to Force=650.

Table 3.2	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
alpha	602.388	18.233	566.781	638.451	0.18	0.118
beta[Hardness]	82.207	14.27	54.61	110.389	0.11	0.076
beta[Rake]	-14.778	13.968	-42.061	12.971	0.089	0.075
beta[Helix]	-146.605	14.736	-175.478	-117.302	0.106	0.081
beta[Relief]	3.915	14.383	-24.507	32.196	0.105	0.08
beta[Corner]	7.962	14.135	-20.117	35.809	0.101	0.082
beta[Flute]	80.331	14.557	51.957	109.538	0.093	0.079
sigma	31.832	5.403	21.943	42.947	0.054	0.033

3c)

Based on the description that the response force is directly linked to the likelihood of missingness, the MNAR model would be the appropriate model to use. In terms of minimising the posterior mean, there a few points to consider. Firstly, this is a linear regression problem where the process variables (x_i) are scaled between 0 and 1, hence at this inflection point we would expect all values to be either 0 or 1. To find the minimum point, any variables with a positive β coefficients would be set to 0 and negative β coefficients would be set to 1 (contributing -1 to the posterior mean).

From results above our minimum posterior mean equation is the following:

$$\min(\mu) = \beta_0 - \beta_1 - \beta_2 + \beta_3 + \beta_4 + \beta_5$$

where the process variables are:

Hardness=0, Rake=1, Helix=1, Relief=0, Corner=0 and Flute=0

3d)

Using the process variable values from above to make a new prediction:

Table 3.3	mean	sd	hdi_2.5%	hdi_97.5%	mcse_mean	mcse_sd
likelihood[0]	441.217	37.09	364.807	511.914	0.206	0.161